

RPCRA Reproducibility Code

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The book Robust Portfolio Construction and Risk Analysis (RPCRA) has a distinctive reproducibility theme, according to which most Figures and Tables in Chapters 2 through 7, and Chapters xx, yy, zz, are reproducible with R scripts and data sets in the PCRA R package companion to the book. This reproducibility theme is conveniently supported by our use of the L^AT_EX (https://www.lyx.org/ document processor to write the RPCRA book, whereby all code to create figures and tables are contained as L^AT_EX document R code chunks that are easily exported in form of single R scripts for each of many chapters.

Although it is nice to be able to use these scripts to recreate the figures and tables of various chapter, this in itself is not the primary purpose of providing the reproducibility code. The main reason for doing so is that it facilitates easy experimentation with different portfolio strategies by easily changing one, or a few, parameters in the scripts, for example changing a smallcap portfolio example in the RPCRA book, to a microcap, or midcap, or largecap portfolio, or changing the number of stocks, in the portfolio, or changing the industry sector of the portfolio, or changing the portfolio optimization constraints or penalties parameters. We believe this considerably enhances a student learning experience, and provides instructors with the ability to easily create portfolio performance and risk homework exercises for their class.

1 R and RStudio

This Vignette assumes that you have already installed both R version 4.4.3 or later, and a current version of RStudio. If you have not already done so, you can find out how to install R and RStudio by reading appropriate parts of Appendix A in *Hands on Programming with R*, available online at <https://rstudio-education.github.io/hopr/>. We also assume that you have used RStudio to install the CRAN current version of the PCRA package, and have already read the Vignette “PCRA Package Overview” that is available at <https://CRAN.R-project.org/package=PCRA>.

2 PCRA Reproducibility Code

The R reproducibility scripts are contained in the demo folder of the PCRA package.

For any package that you have already installed, but have not necessarily loaded, you can get a list of the demo folder R script names by typing the following in the RStudio Console:

```
demo(package = “packageName”)
```

Thus, assuming that you have already installed the PCRA package, typing

```
demo(package = “PCRA”)
```

results in a display of the following list in an RStudio pop-up R demos tab:

Demos in package ‘PCRA’:

Ch2_Foundations_Demo

Computes Chapter 2 figures and tables.

Ch3_ModelingAssetReturns_Demo

Computes Chapter 3 figures and tables.

Ch4_CovarianceMatrices_Demo

Computes Chapter 4 figures and tables.

Ch5_CovarianceMatrices_Demo

Computes Chapter 5 figures and tables.

Ch6_CovarianceMatrices_Demo

Computes Chapter 6 figures and tables.

IntroductionCRSPstocksAndSPGMlfactors

Code for Vignette with same name.

RobustStatisticsForPortfoliosJPM2023

Reproduces figures in JPM 2023 paper.

The code for the demo script of a given chapter has comments to label each figure number and table number in the chapter.

We suggest that you install the `PortfolioAnalytics` package, and have a look at the long list of demos in it.

For any loaded R package, you can view the code for any R script in the demo folder with the command “??” (but **not** with the `help` function!).

For example with the PCRA loaded, use of the command

?? Ch2_Foundations_Demo

results in an RStudio Help tab display with the following two links, separated considerably side by side:

[PCRA::Ch2_Foundations_Demo](#)

[\(Run demo\)](#).

Do **not** Click on the right-hand link, at least not until you have read the NOTE below. Instead, Click on the left-hand link, upon which you will see a display of the entire `Ch2_Foundations_Demo.R` script in the Help tab. Just copy/paste that script into your own new R file, and save it wherever you like. Then in RStudio make the folder where you have saved the demo script the working directory by selecting:

Session > Set Working Directory > To Source File Location

You naturally might want to begin by running the entire Demo script for a given chapter, which you can do by highlighting the entire script and clicking the Run button, and then confirming that the code actually

produces the figures and tables in the RPCRA book chapter. Be forewarned that while many of the code chunks will run quite quickly, e.g., a few seconds, a few of them take longer, sometimes several minutes or more.

After, or instead of, running all the code, you should run the portion of the demo script code at the beginning, where various R packages are loaded, through the first table or figure. You can then search for the code that creates specific figures or tables, and run just those portions of the script.

NOTE: We do not recommend general use of [\(Run demo\)](#). This is because it runs the entire demo script, which will sometimes take a long time. Furthermore, some scripts may fail to execute properly when run this way, and will give you little indication of what the problem is. So if you want to run it all, we recommend doing so as suggested above. That said, the [\(Run demo\)](#) link can be handy for quickly running R demo scripts that execute quickly.

Sourcing Files in RStudio

You can also run an entire script using the R `source` function, with the script name as its argument. But even better, RStudio provides a useful way to run an entire script with a single click – just click on the Source button at the top right corner of the Source Pane. Unfortunately, while this is convenient, it will fail spectacularly if the Source button is pressed without first changing the working directory to the location of the script, and the `rstudioapi` library addresses this problem.

With just two additional lines of code ahead of the start of the script, `rstudioapi` automates the process of setting the working directory to the location of a script when sourcing it. The library and the corresponding lines of code are shown below.

```
library(rstudioapi)
current_path <- rstudioapi::getActiveDocumentContext()$path
setwd(dirname(current_path))
```

This proves most useful when sourcing multiple scripts stored in multiple locations. You may wish to add those lines to your copied/pasted versions of the scripts in the demo folder, so that you can run any of the entire scripts by simply clicking on the RStudio source button.

Character of the Reproducibility Demo R Scripts

The first part of each Chapter’s demo R script contains instructions for installing and loading the R packages that are needed to run all the code in the script. You should follow these instructions carefully.

The R demo scripts are fairly long, with approximately 1000, 900, and 800 lines (including spacing lines) for the Chapters 2, 3, and 4 scripts, respectively. These three scripts require installation and loading of

about a dozen R packages, and each script uses at least one R function, and sometime a number of several functions from each package. In order to help the user be aware of which function belong to which loaded packages, we use the convention “packageName::functionName”.¹ One exception to this practice is in cases where we use a number of functions from the PortfolioAnalytics package, which supports specification of a set of portfolio characteristics such as “full-investment”, “long-only”, “weights constraints”, etc., and out-of-sample back-testing of cumulative gross return and drawdowns. For details on the use of PortfolioAnalytics, we highly recommend the Vignette “An Introduction to Portfolio Optimization with PortfolioAnalytics” by Ross Bennett, which can be downloaded from <https://CRAN.R-project.org/package=PortfolioAnalytics>.

3 Devtools and Rtools

The reproducibility code for some chapters of the RPCRA book will require use of the devtools R package for installing a few R packages, and you might as well install devtools now as follows:

In RStudio drop down Tools menu, select Install Packages, and type devtools in the blank “Packages” field. In the “Install to Library:” field you will then see the entry C:/R/R-4.4.x/library[Default], with $x \geq 3$. Click the “Install” button to initiate the installation of the devtools package.

Eventually, you may want to compile one or more packages from source using a Windows operating system, and in that case you will need to install the Rtools package, which you can do in the same manner as above. Keep in mind that each major version of R (e.g. 4.x) has its corresponding version of Rtools.

¹This device is actually not needed for packages that are loaded, but we believe it none-the-less will be helpful to use it. We note also that use of “packageName::functionName” allows you to use the function with functionName even when the package is only installed, but not loaded.